# The GEDCOM 5.5.5 Specification with Annotations


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**First Release**
**2 Oct 2019**



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This publication, *The GEDCOM 5.5.5 Specification with Annotations* is based on the *The GEDCOM 5.5.1 Specification, Annotated Edition*, which is an annotated edition of the *GEDCOM 5.5.1 Specification*, which is a minor update of the *GEDCOM 5.5 Specification*. The GEDCOM 5.5 and 5.5.1 specifications were created by FamilySearch.

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This publication, *The GEDCOM 5.5.5 Specification with Annotations*, is published for the purposes of review and programming of genealogical software. It is based on the *The GEDCOM 5.5.1 Specification, Annotated Edition*, an in-depth and in-place review (for the purposes of programming), of the *FamilySearch GEDCOM 5.5.1 specification*. This is not only allowed by the GEDCOM 5.5 copyright notice, it is even allowed by the less permissive GEDCOM 5.5.1 copyright notice.

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# Conventions

## Language

This document uses English spelling and *logical quoting*; quotes go around an item, full stops are neither moved into nor out of quoted text.

### must, must not, should, should not

This specification uses *must*, *must not*, *may*, *should* and *should not* as defined in IETF RFC 2119. Thus, other sentences in this specification almost always uses *must not* instead of *may not*. This specification sometimes italicises these phrases for emphasis, but does not capitalise these phrases.

### Records

A GEDCOM files consists of *records*. A record that is contained within another record is a *subrecord*.
Previous versions of GEDCOM have also referred to records and subrecords as *structures* and *substructures*, even “record structures”. They've also been referred to as “tag”, “tag context” and just “context”, while “context” is generally used to refer to the enclosing record, not the subrecord itself.

A GEDCOM record consists of several parts, the *tag* is only part of the record. This version of GEDCOM consciously avoids using “tag” when “record” is meant.
A GEDCOM record starts with a level number. Records with level number zero are known as *top-level records*. Previous GEDCOM versions have referred to top-level records as “zero-level records”, “logical GEDCOM records”, “logical records”, and “record at level zero”; this specification always uses “top-level record”.

The term *records* includes *subrecords*, unless restricted by a modifier, as in “top-level records”, or “INDI records”.

#### Subrecord Notation

The GEDCOM specification uses a convenient subrecord notation to indicate particular subrecords. For example, INDI.NAME is the NAME subrecord of the INDI record, and HEAD.GEDC.VERS is the VERS subrecord of the GEDC subrecord of the HEAD record.

#### Bolded Tags

Tags in GEDCOM syntax definitions are bolded when they are mandatory in that context.

## Characters, Code Units, and Code Points

Characters, code points and code units are related but different concepts. This specification assumes that the reader is familiar with these.

Previous GEDCOM versions expressed fields length in characters. GEDCOM 5.5.5 expresses field length in *code units*.

The text distinguishes between character sets and encodings; GEDCOM 5.5.5 demands the use of Unicode character set, either the UTF-8 or UTF-16 encoding. GEDCOM 5.5.5 continues the use of the HEAD.CHAR line value “UNICODE” (all-uppercase) to indicate UTF-16.

## Terminator versus Newline

This text continues to use “terminator” for the GEDCOM line terminator, but uses “newline” for line breaks in (long) logical line values. This makes discussion of the CONT record easier to understand.

## GEDCOM Readers & Writers

A *GEDCOM writer* is code that writes a particular version (and encoding) of GEDCOM. A *GEDCOM reader* is code that reads a particular version (and encoding) of GEDCOM.
A *GEDCOM 5.5.1 writer* is code that writes GEDCOM 5.5.1 files. A *GEDCOM 5.5.5 reader* is code that reads GEDCOM 5.5.5 files.
A typical genealogical system contains multiple GEDCOM readers and writers, for the last few versions of GEDCOM.

## GEDCOM Examples

The GEDCOM specification contains some GEDCOM examples.
An effort has been made to ensure that all examples are valid GEDCOM, that do not only follow GEDCOM rules and best practice, but genealogical best practices as well.

## Deprecated Features

The GEDCOM 5.5.5 specification contains few deprecated features.

Deprecated sections are highlighted through text & background colour, like this example:

**escape_text:=**

[ alphanum | escape_text + alphanum | escape_text + space ]

where:

space = U+0020, the Space character

The escape_text is the part of an escape between the opening and closing at sign (@). Escape sequences are defined by a GEDCOM form for that GEDCOM form.

GEDCOM 5.5.1 allows almost any Unicode character inside escape sequences.
GEDCOM 5.5.5 restricts escape sequences to alphanumerical characters and the space character.
Notice that escape_text must start with an alphanumerical character, that it must not start with a space.

The space character is allowed but deprecated; it is only included to keep the already existing escape sequence `@#DFRENCH R@` legal.

The general guideline for deprecated features is that they should not be used. Applications should avoid using deprecated features in the GEDCOM files they export.
The general guideline for import of deprecated features is that developers *should* not add support for deprecated features, but should not remove any already existing support either.

# About GEDCOM 5.5.5

## Maintenance Release

Technically, the GEDCOM 5.5.5 specification succeeds the GEDCOM 5.5.1 specification (1999 CE) as the current GEDCOM specification. Practically, it succeeds the *GEDCOM 5.5.1 Specification Annotated Edition* (2019 CE).

Previous GEDCOM releases were feature releases that introduced major new features. GEDCOM 5.5.5 does not introduce any major new features, quite the opposite: GEDCOM 5.5.5 removes long obsolete and deprecated features. GEDCOM 5.5.5 is a maintenance release; it fixes a variety of issues to provide a better specification, and a solid basis for future versions.

The *GEDCOM 5.5.5 Specification with Annotations* is based on the *GEDCOM 5.5.1 Specification Annotated Edition*. It is, despite extensive editing, essentially the same specification, but many textual corrections applied instead of highlighted, obsolete and deprecated parts removed, ambiguities and contradictions resolved. Both the GEDCOM grammar and the Lineage-Linked Form have been simplified without really changing things; whatever isn't allowed anymore wasn't being used anyway, or was deprecated decades ago.

The GEDCOM 5.5.5 focus is on corrections, clarifications, fixes and simplifications, but it's been twenty years since GEDCOM 5.5.1, and plenty of issues remain. The GEDCOM 5.5.5 specification is the GEDCOM 5.5.5 specification *with Annotations*. That isn't the ideal approach, it is the practical approach.

The GEDCOM 5.5.5 is based on the *GEDCOM 5.5.1 Annotated Edition* and contains some of the same annotations as the *Annotated Edition*.
Many annotations have been applied, replaced by a better specification, replaced by new sections or turned into bonus chapters, but yet other annotations remain.
The annotations and bonus chapter provide useful and interesting information, but are *not* part of the specification proper.

## Quality

The GEDCOM 5.5.5 specification is about GEDCOM quality. GEDCOM 5.5.5 is a better GEDCOM; The GEDCOM 5.5.5 specification is a better specification, and GEDCOM 5.5.5 files are better GEDCOM files. The GEDCOM 5.5.1 and 5.5.5 specification are so close, that most high-quality GEDCOM 5.5.1 files are GEDCOM 5.5.5 files in all but version number. The GEDCOM 5.5.5 version number is badge of quality.

Developers of products that already produce GEDCOM 5.5.1 files that already conform with all GEDCOM rules, recommendations and best practices documented in the *GEDCOM 5.5.1 Annotated Edition* (2018 CE) will find it easy to upgrade the GEDCOM output of their products to GEDCOM 5.5.5.

GEDCOM 5.5.5 simplifies and tightens the rules to spark a change in genealogy industry culture. There are technical differences between GEDCOM 5.5.1 and GEDCOM 5.5.5, but the most important change is that GEDCOM 5.5.5 demands strict compliance.
Many existing GEDCOM 5.5.1 readers tolerate a wide variety of errors known to occur in GEDCOM 5.5.1 files,. GEDCOM 5.5.5 readers *need not* accepts errors known to occur in GEDCOM 5.5.1 files, and *must not* do so. GEDCOM 5.5.5 readers must make a clean break with such practices. Developers improving the GEDCOM output of their products should continue to use version number 5.5.1 until their GEDCOM output is in compliance with GEDCOM 5.5.5.

It is perfectly possible to combine a GEDCOM 5.5 and GEDCOM 5.5.1 reader into a single GEDCOM 5.5.0/1 reader, it is not possible to support GEDCOM 5.5.1 and 5.5.5 with a single combined reader.
A GEDCOM 5.5.5 reader must be separate from readers for GEDCOM 5.5.1 and earlier, not just to make sure it doesn't support obsolete Lineage-Linked Form records, but because the GEDCOM 5.5.5 grammar is considerably simpler and stricter than the GEDCOM 5.5.0/1 grammar. A GEDCOM 5.5.5 reader may use some of the same building blocks as the GEDCOM

5.5.1 reader, but it *can not* and *must not* be a combined GEDCOM 5.5.1 & 5.5.5 reader.
The GEDCOM 5.5.5 reader *need not* and *must not* compensate for errors that are only known to occur in ostensible GEDCOM 5.5.1 and earlier files, and *must* keep things simple by rejecting invalid files.
A GEDCOM 5.5.5 reader should be very clean code that is significantly simpler and noticeably faster than older GEDCOM readers.

Developers should keep their existing readers for GEDCOM 5.5.1 and earlier for backward compatibility, but are advised take advantage of the GEDCOM 5.5.5 specification to improve their GEDCOM 5.5.1 writer, and bring its output as close to that of the GEDCOM 5.5.5 writer as possible.

The *What's New in GEDCOM 5.5.5* section provides an overview of changes, including such things as CONC and CONT rules, tables of superfluous and obsolete records, and is followed by *GEDCOM writer & reader checklists*.

# What's New in GEDCOM 5.5.5

This section provides an fairly detailed overview of changes between GEDCOM 5.5.1 and GEDCOM 5.5.5. The GEDCOM 5.5.5 specification does not include a overview of changes before GEDCOM 5.5.1. Those changes happened more than two decades ago, and are listed in the *GEDCOM 5.5.1 Annotated Edition*.

GEDCOM 5.5.5 isn't a feature release, it's a maintenance release; while a few small additions have been sneaked in, but there are no major new features.

Both the GEDCOM Grammar and the Lineage-Linked Form have been simplified. A major improvement is the introduction of a Basic GEDCOM Language and the top-level <GEDCOM_FILE> definition.
Text has been improved with current and more accurate terminology. When there were two ways to do the same thing, there's one way to do that thing now. Contradictions have been resolved, existing specifications has been clarified. Obsolete, deprecated and superfluous stuff has been taken out.

## General Improvements

### New Specification Parts

- A short *Conventions* section introduces some terminology and terminology changes.
- A section on CONC & CONT has been added to Chapter 1
- That new section on CONC & CONT is preceded by a new section about white space
- New sections defining the Basic GEDCOM Language
- The chapter on Character Set & Encodings has been rewritten.
- New chapter on system identifiers.
- GEDCOM Reader & Writer Checklists
- New appendix offers consolidated ANSEL / Unicode conversion tables (was bonus chapter in the Annotated Edition)
- Bonus chapters on a variety of issues, including GEDCOM Validation, GEDCOM version detection, and GEDCOM 5.5.1 version detection. The new bonus chapter on GEDCOM Version detection contains some advice on how to support GEDCOM 5.5.5 while continuing to support GEDCOM 5.5.1.

### Textual Corrections

#### Corrections

Textual improvements include corrections of spelling errors, editing errors and conceptual errors, and improved terminology.
There are many small changes, some more important than others, but the many small changes really all add up to a significantly clearer, more consistent and more accurate specification.

#### English

As this is an international standard, an effort was made to use International English throughout.

#### Logical Quoting

Switched from illogical to logical quoting style to avoid confusion and misinterpretation; logical quoting style leaves no doubt what's quoted and what's not.

**Terminology**

The specification now uses better, more current, more accurate and more consistent terminology throughout. The text now uses “file” instead of “transmission”, “transcription” instead of “variation” and properly distinguishes between “tag” and “record” by no longer using “tag” when “record” is meant.
There is no more conflation of tag name and meaning into a mixed-case word such as “INDIvidual” either.

| old | new |
|---|---|
| zero level record | top-level record |
| logical record | top-level record |
| tag | record |
| tag context | record |
| context | record |
| family record | family group record |
| transmission | file |
| variation | transcription |
| reading | transcription |
| biological | official |
| DESTination | DEST (destination) |
| MARRiage | MAR (relationship) |
| UNICODE | Unicode |
| UNICODE | UTF-16 |
| characters | code units |
| sent the data | created the file |
| 8-bit ASCII | (contradiction in terms) |
| ASCII (USA Version) | ASCII |
| record structures | records |
| (wide) characters | code units |

**Official versus Biological**

The text of the specification no longer assumes that recorded relationship are biological. Relationships recorded on the basis of official paper documentation are official, and should not be assumed to be biological.

**Family versus Family Group**

The text is no longer written as if *family* and *family group* are the same concept. A so-called *family group* is *one couple and their children*. A single family often involves more than one such group.

**Better Record Names**

The FAM record is no longer referred to as the “family record”, but more accurately as the “family group record”. Similarly, the MARR record, which is not only used to record marriages, but for recording every relationship type between two people, is no longer referred to as the marriage record, but as the relationship record.

**Appendix A Definitions**

Significantly improved Appendix A definitions; many definitions are more objective, clearer and more accurate now.

# Significant Simplification

## One file per GEDCOM File

Multi-volume GEDCOM files (defined in GEDCOM 5.5.1 chapter 2) are no longer legal; a GEDCOM file must be just one file now. The multi-volume feature was obsolete before it was introduced, and remains largely unsupported. With GEDCOM 5.5.5 it is one file per GEDCOM file.

## One GEDCOM Version per File

The GEDCOM 5.5.1 specification contains the demand to support a particular GEDCOM 5.5 record, while that record is not legal GEDCOM 5.5.1. The GEDCOM 5.5.1 <<MULTIMEDIA_LINK>> definition demands that applications support the GEDCOM 5.5 <<MULTIMEDIA_LINK>> record as well, because “some applications” use GEDCOM 5.5.1, yet the GEDCOM 5.5 <<MULTIMEDIA_LINK>> record.

GEDCOM does not demand anything like that. On contrary, GEDCOM 5.5.5 does not allow it; a GEDCOM form cannot make such a demand. From GEDCOM 5.5.5 onwards, it is one GEDCOM version per file; a GEDCOM 5.5.5 file must contain GEDCOM 5.5.5 records. A GEDCOM 5.5.5 reader must not tolerate anything that's only legal in another GEDCOM version.

## One line terminator per GEDCOM file

The GEDCOM 5.5.5 definition of the line terminator makes is clear that the line terminator that ends each GEDCOM line must be the same for all GEDCOM lines in a GEDCOM file.

## One Encoding per GEDCOM File

GEDCOM 5.4, 5.5 and 5.5.1 contains an *ANSEL demand*; the encoding of the GEDCOM header must use ANSEL, regardless of the encoding that header specifies for the (rest of) the GEDCOM file GEDCOM 5.5.5 demands that the encoding specified in the GEDCOM header be used for the entire file, including the header itself.

## One Character Set for GEDCOM

GEDCOM 5.5.5 no longer allows ASCII or ANSEL. GEDCOM 5.5.5 is Unicode-only.

## One Way of Doing Things

Where GEDCOM 5.5.1 still offers two ways of doing things, GEDCOM 5.5.5 offers one way. These changes have been a long time coming; GEDCOM 5.5.5 is only finishing transitions that FamilySearch started in GEDCOM 5.4 and 5.5 (1995 CE). Developers and users have had more than a quarter century to adopt the changes made by FamilySearch.

FamilySearch added structured addresses and deprecated unstructured addresses in GEDCOM 5.5. GEDCOM 5.5.5 demands structured addresses only.
Similarly, FamilySearch introduced the citation-source-repository model in GEDCOM 5.4, and deprecated the unstructured free-text citations. FamilySearch even gave instructions for providing structure in the previously free-form text field. GEDCOM 5.5.5 demands the use of source and repository records.

# GEDCOM Grammar

There are clarifications, corrections, improvements, simplifications and more.

## One file per GEDCOM file

The multi-volume support still demanded by GEDCOM 5.5.1 is a significant burden on GEDCOM readers; GEDCOM 5.5.1 readers must contain file handling logic to find and load multiple volumes from removable media, and prompt the user for a media change when a volume is missing. Its removal is a significant simplification of GEDCOM.

## Text Files

The specification now explicitly states that GEDCOM files are text files, and that while GEDCOM allows multiple line terminators, each GEDCOM file must use a single line terminator choice throughout the file.

### CR, LF & CR/LF

GEDCOM 5.5.5 continues to allow CR, LF & CR/LF as line terminators, but no longer allows LF/CR as a line terminator. This is a correction of a mistake; no platform uses LF/CR, and its inclusion creates problems for GEDCOM readers.
This correction is a real change to the GEDCOM grammar, yet does not affect any users; the LF/CR terminator only occurs in GEDCOM 5.5 and 5.5.1 test files specifically created to test support for it...

## Tags Case-Sensitive

This version of the specification clearly states that GEDCOM tags are case-sensitive.

## Legal Characters

The GEDCOM grammar has been edited to clearly separate which characters are legal in tags, identifiers and escapes sequences, and which characters are legal as line values.

### Tags Underscore

Tags remain restricted to alphanumerical characters and the underscore. The tag definition has been tightened; the underscore may only occur as the first character now, and may no longer be the only character.
Incidentally, it was and remains legal for a tag to start with a digit.

### Identifiers must be Alphanumerical

The GEDCOM 5.5.1 pointer definition allowed many characters inside cross-reference identifiers, even spaces, something that is almost sure to trip up a parser. It even allowed the bulk of all Unicode characters, thus creating more issues for a GEDCOM parser to deal with. In the real world, the flexibility provided by this definition was only used in GEDCOM files specifically created to test the ability of GEDCOM readers to handle this flexibility.

The GEDCOM 5.5.5 specification restricts the cross-reference identifier definition to alphanumerical characters, to allow straightforward parsing.
This allows GEDCOM 5.5.5 readers to be significantly simpler than GEDCOM 5.5.1 readers.

### Escape Sequences

The updated GEDCOM grammar no longer allows almost any character in escape sequences. Escape sequences are restricted to alphanumerical characters, now except that spaces are still allowed. Spaces are allowed but deprecated because the Lineage-Linked Grammar defines one escape sequence with a space in it.

**Tab is Legal**

Use of Horizontal Tab in user text is legal now. Most genealogy applications already allowed the use of tabs in text, despite the GEDCOM 5.5.1 specifications disallowing it. GEDCOM 5.5.5 makes the existing practice legal.

**Significant Tag Length**

The GEDCOM grammar allows tags to be 31 code units long. The GEDCOM 5.5.1 specification adds that the first 15 of those 31 must be unique. What this really means is that while tags may be longer, their so-called *significant length* is only 15 code units. The practical consequence is that a GEDCOM reader may ignore everything beyond the 15th code unit, and that may result in two different identifiers being interpreted as being the same one.
This rule has never been off much practical consequence as all the standard GEDCOM tags as well as those of the Lineage-Linked Form are considerably shorter. Thus, this rule only affect GEDCOM extensions.
Nevertheless, GEDCOM 5.5.5 ends the complexity created by the limited significant length rule: the entire tag is significant now.

**Maximum Record Size**

The GEDCOM grammar in GEDCOM 5.5.1 limits the size of top-level records; they should fit in a memory buffer of less than 32K. That restriction made sense in the 1980s, it is obsolete now.
The GEDCOM grammar in the GEDCOM 5.5.5 specification no longer limits the size of top-level GEDCOM records.
Some records within the Lineage-Linked Form, which did not have an explicit maximum length before, have had their maximum length set at 32.765 code units.

**line_item Fixed**

The FamilySearch GEDCOM 5.5.1 specification defined line_item like this:

```
[ any_char | escape | line_item + any_char | line_item + escape ]
```

According to that recursive definition:

- line_item may consist of nothing but a single escape sequence.
- line_item may consist of a some text followed by an escape sequence.
- line_item may consist of a some text followed by multiple escape sequence.
- line_item may consist of a some text both preceded and followed by an escape sequence.
- line_item may consist of nothing but multiple escape sequences.

Most of that is not as intended. Most of these possibilities are wrong.

The GEDCOM 5.5.5 line_item definition explicitly restricts line_item to just three different formats:

- just text
- just one escape sequence
- just one escape sequence followed by text

**pointer is an xref_ID**

GEDCOM 5.5.1 defines xref_ID as a pointer. That's the wrong way round. GEDCOM 5.5.5 defines a pointer as an xref_ID. As part of this change, pointer_string has been renamed identifier_string.

#### White Space Rules

The new *White Space* section adds white space rules:

- **GEDCOM lines *must not* not end with superfluous trailing white space.**
- **GEDCOM lines *may* contain significant trailing white space.**
- **A GEDCOM reader *must* import every line value as-is, *must not* not strip trailing white space.**
- **A GEDCOM reader *must* import every line value as-is, *must not* not strip leading white space.**

#### New CONC & CONT Rule, Clarification and Guideline

- Rule: The CONC & CONT records *must not* be used with records defined in the GEDCOM grammar. They may only be used with records defined in a GEDCOM form.
- Clarification: It is illegal to have CONC or CONT records subordinate to CONC or CONT records.
- Guideline: GEDCOM writers *should not* use CONC records for line values with a maximum length of 248 code unit or less.

#### Simple CONC & CONT Processing Rules

The new section on CONC & CONT ends with CONC & CONT processing rules for GEDCOM writers & readers to ensure correct transfer of white space. Those rules are the simplest yet, literally *requiring no effort* from GEDCOM readers at all; a GEDCOM reader must import all lines *as-is*.

## Character Sets & Encodings

### Unicode

Unicode replaces ANSEL as *the* GEDCOM character set. GEDCOM 5.5.5 does no longer allow ASCII or ANSEL. GEDCOM 5.5.5 demands Unicode. The recommended encoding is UTF-8+ for Western applications, UTF-16 for Eastern applications. Applications, when asked to create a GEDCOM file, need not offer users a choice between these two.

Import of either encoding should never be an issue. Applications *must* be able to import both UTF-8 and UTF-16 GEDCOM 5.5.5 files.
Adding UTF-16 GEDCOM import to a Unicode application that already support UTF-8 GEDCOM import is fairly easy to do, as the application is using UTF-16 internally. The import only needs to pay attention to the normal form, and the application can rely on operating system routines for that.

### Unicode Applications Only

GEDCOM 5.5.5 is for Unicode applications. Legacy applications that are still code page-based *must not* use GEDCOM 5.5.5, but must continue to use GEDCOM 5.5.1. Code page applications can and should follow all the rules, recommendations and best practices provided the *GEDCOM 5.5.1 Annotated Edition*.

#### Byte Order Mark Mandatory

Unicode demands Byte Order Mark on UTF-16 files, allows them on UTF-8 files. GEDCOM 5.5.5 keeps it simple: use of a Byte Order Mark (BOM) on UTF-8 GEDCOM files is mandatory for all GEDCOM 5.5.5 files, both UTF-8 and UTF-16 GEDCOM files. This simplifies GEDCOM detection considerably. Systems reading GEDCOM 5.5.5 files know the character set and encoding used in the header as soon as they've processed the Byte Order Mark (BOM), *before*

they start reading the header.

#### Unicode Code Point Notation

The specification now uses Unicode point notation to indicate specific characters; e.g. U+0020 is the Unicode code point for the Space character.

### ANSEL

Starting with GEDCOM 5.5.5, the ASCII and ANSEL character sets are illegal. It is illegal to use ANSEL in GEDCOM 5.5.5, so a system need not support ANSEL at all to be fully GEDCOM 5.5.5-compliant. Systems should support ANSEL for import of ANSEL-encoded GEDCOM 5.5.1 and earlier files.

#### Consolidated ANSEL Table

A single consolidated ANSEL table solves the problem of different “ANSEL” tables in different GEDCOM versions. This single consolidated table can and should be use for all older GEDCOM versions; that's why it's there.

#### Strict Conformance

GEDCOM 5.5.5 does not tolerate the use of illegal characters sets or encodings.

- GEDCOM 5.5.5 writers *must not* use anything but the legal encodings.
- GEDCOM 5.5.5 readers *must not* accept anything but the legal encodings.
- GEDCOM 5.5.5 readers *must* reject all illegal encodings.

## GEDCOM Language

### Basic GEDCOM Language

The GEDCOM 5.5.1 *Chapter 1 Data Representation Grammar* defines just two records types: CONC & CONT. The HEAD & TRLR record are not even mentioned.
The chapter has been extensively rewritten and extended to define the Basic GEDCOM Language, including the HEAD & TRLR records.

### Multiple GEDCOM Forms Problem

The GEDCOM 5.5.1 specification allows multiple GEDCOM forms, yet it doesn't...
GEDCOM 5.5.1 Chapter 2 defines the Lineage-Linked Form, and that includes the definition of the GEDCOM header, which specifies the GEDCOM form being used. Specifying a GEDCOM form within the Lineage-Linked GEDCOM header does not make sense, as that entire GEDCOM header is part of the Lineage-Linked Form.
The HEAD.GEDC.FORM line value supposedly specifies the GEDCOM form, but `LINEAGE-LINKED` is the only legal value, and that suggests that the HEAD.GEDC.FORM record is superfluous and redundant.

### Basic GEDCOM Language Solution

The GEDCOM 5.5.5 specification distinguishes between the Basic GEDCOM Language and the Lineage-Lineage Link Form. The record that specifies the GEDCOM form is no longer part of a specific GEDCOM form, but part of the Basic GEDCOM Language.
GEDCOM 5.5.5 defines a Basic GEDCOM Language first, and only then defines the Lineage-Linked Form. The Basic GEDCOM language defines the GEDCOM header and trailer record in addition to the CONC & CONT records. The HEAD.GEDC.FORM is part of the Basic

GEDCOM Language now, and really specifies which GEDCOM form is being used.

**GEDCOM Header Split**

The old GEDCOM header has been split into a basic GEDCOM and an extension. The Basic GEDCOM header is defined by the Basic GEDCOM Language, the extension is defined by the Lineage-Linked Form.

**Trailer Record**

In GEDCOM 5.5.1, the TRLR record signifies the end of the GEDCOM file.
However, in GEDCOM 5.4, the TRLR record may be followed by embedded multi-media objects, with the real end of the GEDCOM file marked by an the END record.
GEDCOM 5.5 still supports embedded multimedia records, but they appear before the TRLR record, just like all other records.

By making the header and trailer records part of the Basic GEDCOM language, and demanding that all GEDCOM form records occur in between, the GEDCOM 5.5.5 specification ensures that the trailer record is truly the trailer record.

# CONC & CONT

## CONC & CONT Section

The *GEDCOM 5.5.1 Annotated Edition* collected information and instructions about CONC & CONT together in a large annotation. The GEDCOM 5.5.5 specification provides a new *CONC & CONT* section as part of Chapter 1.

## CONC & CONT Restriction

The GEDCOM specification allows CONC & CONT anywhere they are needed. GEDCOM 5.5.5 introduces a restriction, that should have been part of GEDCOM from the beginning, to keep parsing GEDCOM files simple; CONC & CONT must not be used with basic GEDCOM records, only with records defined by a GEDCOM form.

It is illegal to use CONC or CONT within the Basic GEDCOM header. It is legal, but strongly discommended, to use CONC or CONT within a form-specific header extension. It is important to keep the GEDCOM header simple. The GEDCOM header is not only examined by GEDCOM readers, but also by general utilities that report file types.

## New CONC Guideline

GEDCOM 5.5.5 introduces a new guideline for CONC usage; GEDCOM writers should not use CONC for line values with a maximum length of 248 code units or less.
A GEDCOM reader must not complain about unnecessary use of CONC records as long as they're valid, but a GEDCOM validator should give a warning.

## Simple CONC & CONT Rules

Historically, CONC & CONT handling has been problematic. The *GEDCOM 5.5.1 Annotated Edition* provided simple guidelines for getting it right.
The GEDCOM 5.5.5 specification provides a new section about white space, and then provides the simplest rules possible:

- A GEDCOM writer must not write superfluous white space, may only write significant white space.
- A GEDCOM reader must treat all white space as significant; it may never trim leading or

trailing white space from lines values.

Those two simple rules ensure correct transfer of long text files, even for pathological cases such as long stretches of white space.

# GEDCOM Header

#### Simplified GEDCOM Detection

All GEDCOM 5.5.5 files are Unicode files, and all GEDCOM 5.5.5 files, both UTF-8 and UTF-16 files, start with a Byte Order Mark (BOM). The mandatory Byte Order Mark simplifies GEDCOM detection and header interpretation. Systems reading GEDCOM 5.5.5 files know the character set and encoding used in the header as soon as they've processed the Byte Order Mark (BOM), that is *before* they start reading the header.

### Strict Conformance

GEDCOM 5.5.5 demands a correct GEDCOM header. A GEDCOM 5.5.5 reader *must not* process an ostensible GEDCOM file if the header is invalid.

# GEDCOM Header Split

The GEDCOM header defined in GEDCOM 5.5.1 contains both general subrecords as well as subrecords specific to Lineage-Linked Form. To continue to accommodate both categories in GEDCOM 5.5.5 with just one GEDCOM header, the GEDCOM header has been split into a Basic GEDCOM header and a Lineage-Linked Form extension.
The Basic GEDCOM Language defines the Basic GEDCOM header and allows GEDCOM forms to extend it with form-specific records. The Lineage-Linked Form extends the GEDCOM header with Lineage-Linked specific subrecords.

### GEDCOM Recognition

The definition of a Basic GEDCOM header moves all the essential header subrecords to the very front of the GEDCOM file. This aids recognition of GEDCOM files.

### HEAD.GEDC.FORM.VERS

Historically, there is just one version number for both the Basic GEDCOM language and the Lineage-Linked Form.
GEDCOM 5.5.5 adds HEAD.GEDC.FORM.VERS; a version number for the GEDCOM form used, distinct from the GEDCOM version number.

### CONC & CONT

That CONC & CONT must not be used with basic GEDCOM records, only with record defined by a GEDCOM form, specifically means that CONC & CONT must not be used in the <<GEDCOM_HEADER>>.
Use of CONC & CONT is allowed but strongly deprecated for header subrecords defined by a GEDCOM form.

### HEAD.DATE.COPR and HEAD.NOTE

To avoid needing CONC & CONT with header subrecords of the Lineage-Linked Form, the maximum length of the HEAD.DATE.COPR and HEAD.NOTE lines values have both been set at 248 code units.
The 248 code units allowed for <COPYRIGHT_GEDCOM_FILE> is significantly more than the

maximum of 90 code units allowed by GEDCOM 5.4 and 5.5. The 248 code units allowed for <GEDCOM_CONTENT_DESCRIPTION> is more than the maximum 213 code units allowed by GEDCOM 5.4, and identical to the maximum of 248 code units allowed by GEDCOM 5.5.

**GEDCOM Dialect**

GEDCOM 5.5.5 clearly states that it is the HEAD.DEST line value (*not* the HEAD.SOUR line value) that specifies the GEDCOM dialect used, i.e. the interpretation of GEDCOM extensions.

**HEAD.DEST Value Rules**

- GEDCOM 5.5.5 clearly states that the HEAD.DEST line value should default to being equal to the HEAD.SOUR line value.
- GEDCOM 5.5.5 adds that it is illegal for the HEAD.DEST to be a nonsense value.

**HEAD.SUBM**

GEDCOM 5.5.5 allows just one SUBM record for the entire file, and it must come directly after the GEDCOM header. That makes the HEAD.SUBM pointer superfluous. The HEAD.SUBM record is no longer mandatory, and has been deprecated.

**XREF.SUBM**

Obviously, the XREF:SUBM pointer has been deprecated too.

# Lineage-Linked Form

## removal of at-signs

The many at-signs (@), which used to be so characteristic of Lineage-Linked Form definitions, are gone from those definitions. Lineage-linked form definitions used fragments such as `@<XREF:INDI>@`, always surrounding a cross-reference with the at-signs that start and end a cross-reference. The problem with that is not only that it is odd for the record using a cross-reference to demand at-signs around a cross-reference, but that the GEDCOM grammar makes it clear that *the at-signs are already part of the cross-reference*. Thus, those definitions were actually demanding double at-signs.

## Renamed

Some parts of the Lineage-Linked Form have been renamed to ease understanding, avoid confusion and enhance readability:

- <<RECORD>> to <<LINEAGE_LINKED_RECORD>>
- <<FAM_RECORD>> to <<FAM_GROUP_RECORD>>
- <APPROVED_SYSTEM_IDENTIFIER> to <SYSTEM_IDENTIFIER>
- <CHARACTER_SET> to <CHARACTER_ENCODING>
- <FILE_NAME> to <GEDCOM_FILE_NAME>
- <TRANSMISSION_DATE> to <FILE_CREATION_DATE>
- <NAME_PHONETIC_VARIATION> to <NAME_PHONETIC>
- <PLACE_PHONETIC_VARIATION> to <PLACE_PHONETIC>
- <PHONETIC_TYPE> to <PHONETISATION_METHOD>
- <NAME_ROMANIZED_VARIATION> to <NAME_ROMANISED>
- <PLACE_ROMANIZED_VARIATION> to <PLACE_ROMANISED>
- <ROMANIZED_TYPE> to <ROMANISATION_METHOD>
- <NATIONAL_ID_NUMBER> to <ID_NUMBER>

- <SUBMITTER_TEXT> to <USER_TEXT>
- <COUNT_OF_MARRIAGES> to <NUMBER_OF_RELATIONSHIPS>

## Maximum GEDCOM File Length

The maximum length of <GEDCOM_FILE_NAME.SUBM> has been increased from an arbitrary 90 to 248 code units - the maximum length a single GEDCOM line will allow.

## NEW_TAG

There is no tag called NEW_TAG and instructions concerning user-defined record should not be hidden among the definition of GEDCOM form records. There is a small *User-Defined Records* section now.

## Duplicate Records

The GEDCOM 5.5.1 Lineage-Linked From defines several records duplicate functionality of other already existing records. The GEDCOM 5.5.1 Annotated Edition (2018 CE) marked these records as deprecate. GEDCOM 5.5.5 simplifies GEDCOM by removing the duplicate records. The following table summarises which records were still legal in GEDCOM 5.5.1, and what should be used in GEDCOM 5.5.5 instead:

| allowed in GEDCOM 5.5.1 | GEDCOM 5.5.5 |
|---|---|
| INDI.BAPL | INDI.BAPM |
| INDI.BLES | INDI.EVENT |
| INDI.CONL | INDI.CONF |
| INDI.ENDL | INDI.EVENT |
| INDI.ORDN | INDI.EVENT |
| INDI.SLGC | INDI.EVENT |
| INDI.SSN | INDI.IDNO |
| FAM.SLGS | FAM.EVENT |

### LDS Events

The above table contains LDS-specific records. Removal of the LDS-specific records improves support for these LDS events.
Few genealogy applications support the LDS-specific records. On GEDCOM export from those applications and subsequent import into another application, the data recorded in these events are lost. By requiring the use of the universally supported regular baptism record instead of a poorly supported LDS-specific baptism record, GEDCOM 5.5.5 ensures that the recorded information is no longer lost in transfer.

## Obsolete Records

The following obsolete and superfluous GEDCOM 5.5.1 records have been retired, and do not have GEDCOM 5.5.5 alternatives.

| allowed in GEDCOM 5.5.1 | brief comment |
|---|---|
| HEAD.CHAR.VERS | Should never have been included. Never used, only abused. |
| HEAD.PLAC<br>HEAD.PLAC.FORM | Unnecessary complexity. Widely unsupported. |
| HEAD.SUBN | Specific to Ancestral File; obsolete. |
| EVEN.RESN | Specific to Ancestral File; obsolete. |
| FAM.RESN | Specific to Ancestral File; obsolete. |
| FAMC.STAT | Never used, unsupported not needed. |
| FAM.SUBM | GEDCOM 5.5.5 has one submitter per GEDCOM file. |
| INDI.RESN | Specific to Ancestral File; obsolete. |
| INDI.SUBM | GEDCOM 5.5.5 has one submitter per GEDCOM file. |
| INDI.AFN | Specific to Ancestral File; obsolete<br>can use IDNO to maintain but should not. |
| INDI.ALIA | Never used in practice.<br>ASSO remains supported. |
| INDI.ANCI | Never used, not needed. |
| INDI.DESI | Never used, not needed. |
| INDI.RFN | Specific to Ancestral File; obsolete<br>can use IDNO to maintain but should not. |
| SUBM.LANG | Serves no purpose. Already have HEAD.LANG. |
| SUBM.RFN | Specific to Ancestral File; obsolete. |
| SUBN | Specific to Ancestral File; obsolete. |

## <CHILD_LINKAGE_STATUS>

The <CHILD_LINKAGE_STATUS> is not supported by any major applications, and quite possibly by no application at all. It has been marked obsolete in the *GEDCOM 5.5.1 Annotated Edition*, and removed from GEDCOM 5.5.5.
As a result, the STAT tag is obsolete too, no longer used and has been removed from Appendix A.

## Tags obsoleted in GEDCOM 5.5.5

The following tags, legal in GEDCOM 5.5.1, do not occur in GEDCOM 5.5.5 files: AFN, ALIA, ANCI, BAPL, CONL, DESI, ENDL, ORDN, SLGC, RESN, SSN, SLGS, SUBN, and STAT.

## Maximum Identifier Length

Contradiction between the GEDCOM grammar and the Lineage-Linked Form concerning the maximum length of cross-reference identifiers has been resolved. The maximum length is as specified by the GEDCOM grammar.

## C-style Comments

C-style comments in definitions and GEDCOM examples were not universally understood as comments. When they were recognised as comments, they created the mistaken impression that GEDCOM files may contain comments.
GEDCOM does not support comments. Removed all C-style comments from definitions.

## Confusing CONC & CONT

The GEDCOM grammar makes it clear that CONC & CONT are allowed anywhere they are needed. The explicit inclusion of CONC & CONT in the Lineage-Linked Form is an unnecessary complication of the syntax, that encouraged misinterpretation, as explicit inclusion at a few select points in the syntax strongly suggests is not allowed elsewhere...
GEDCOM 5.5.5 respects that this is implicit in any GEDCOM from, and no longer confuses by including CONC or CONT explicitly. All explicit CONC & CONT references have been removed

from the Lineage-Linked Form definition. The definitions in the Lineage-Linked Form are all about the Lineage-Linked Form now, as they should be.

The GEDCOM 5.5.1 specification did not set maximum length for the records concerned. The GEDCOM 5.5.5 specification sets their maximum length at either 248, 4095 or 32.767 (2^15-1) code units. This is in deference to the old (GEDCOM 5.5.1) rule that GEDCOM records should fit into a memory of less than 32KB. A Unicode application using UTF-16 internally will need no more than a 64 KB buffer to hold the largest legal values.

| item | maximum |
|---|---|
| IND.DSCR <PHYSICAL_DESCRIPTION> | 4095 |
| SOUR.AUTH <SOURCE_ORIGINATOR> | 248 |
| SOUR.TITL <SOURCE_DESCRIPTIVE_TITLE> | 4095 |
| SOUR.PUBL <SOURCE_PUBLICATION_FACTS> | 4095 |
| SOUR.DATA.TEXT <TEXT_FROM_SOURCE> | 32.767 |
| SOUR.TEXT <TEXT_FROM_SOURCE> | 32.767 |
| NOTE <USER_TEXT> | 32.767 |

## Same-Sex Marriage

GEDCOM 5.5.5 supports same-sex relationships. Technically, this is not a new feature, but merely a clarification in support of already existing practice. As the *Annotated Edition* already noted, the GEDCOM 5.5.1 specification does *not* really exclude same-sex marriage, but actually supports it *as-is*, i.e. without resorting to any GEDCOM extension. The Lineage-Linked Form specification for recording relationships in the GEDCOM 5.5.5 specification has not changed from GEDCOM 5.5.1. There's no change in syntax, only improved descriptions.

## Sex specified by INDI.SEX

GEDCOM 5.5.5 clearly states the following two rules, already present in the Annotated Edition:

- No assumption about gender should be made based on the usage of the FAM.HUSB or FAM.WIFE records.
- An individual's sex is specified by INDI.SEX, and by INDI.SEX only.

## relationships

The FAM.MARR record documents the relationship between FAM.HUSB or FAM.WIFE. The nature of the relationship is documented by the optional MARR.TYPE subrecord, and GEDCOM 5.5.5 provides a list of the common MARR.TYPE values and their meaning in the definition of the <<FAM_GROUP_RECORD>>.

| value | description |
|---|---|
| unknown | relationship (type unknown) |
| marriage | marriage |
| not married | not married |
| civil | civil marriage |
| religious | religious marriage |
| common law | common law marriage |
| partnership | partnership |
| registered partnership | registered partnership |
| living together | living together |
| living apart together | living apart together |